Predicting Black Carbon Concentrations at SMEAR III, Helsinki, using Random Forest

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1 Introduction

Black carbon (BC) is a primary combustion aerosol whose concentration in Helsinki is shaped simultaneously by traffic and heating emissions, and by the strongly seasonal dynamics of the atmospheric boundary layer. This study trains a **Random Forest** (RF) regressor [1] on hourly measurements from the **SMEAR III** urban station in Kumpula [2], covering 1 September to 31 October 2025, using gas-phase chemistry, meteorology, solar radiation, eddy-covariance heat flux, and rainfall as predictors.

The analysis is as much about methodology as about the result itself. Beginning from a baseline that falls into a **surrogate trap** (CO capturing $> 80\%$ of variance and masking atmospheric physics), the study documents three successive discoveries: (1) a BC/CO ratio target and physics-engineered features raise honest test R^2 from 0.34 to 0.59; (2) naïvely adding true BC lag features inflates R^2 to 0.81, but this is **test-set leakage** ($\Delta R^2_{\text{leak}} = +0.23$), exposed by a walk-forward recursive evaluation; and (3) restricting lags to exogenous CO and NO_x observables recovers genuine atmospheric memory without lookahead, yielding a final honest $R^2 = 0.61$ — a ceiling attributable not to algorithmic failure but to a **fundamental covariate shift** between Helsinki's September and late-October polar-night regimes. Source code available at https://github.com/lixruize-del/ATMCE-M313-Individual-Programming-Exercise.

2 Data and Methodology

2.1 Datasets

All data were downloaded from the SMEAR III station via Smart-SMEAR [2] (1-h averages, arithmetic). Short outages (≤ 3 h) were repaired by linear interpolation; longer gaps and physically impossible values (BC, CO, $NO_x \leq 0$) were removed, yielding **1,401 hourly observations**. Table 1 summarises the six datasets used.

Table 1: Datasets used in this study (Sep 1 – Oct 31, 2025).

Dataset	Variables	Role
Black carbon	BC concentration (ng cm^{-3}, MAAP)	Response y
Gases	NO_x, SO_2, O_3, CO	Predictor
Meteorology	Pressure, wind direction (32 m), T, RH	Predictor
Radiation	Global radiation (W m^{-2})	Bonus predictor
Eddy flux	Sensible heat flux H (W m^{-2})	Bonus predictor
Rainfall	Precipitation intensity (mm h^{-1}, $\times 3$ gauges)	Bonus predictor

2.2 Feature Engineering

Thirty-one predictors were engineered from the raw measurements. Table 2 groups them by physical category.

Table 2: Engineered feature categories and their physical motivation.

Category	Features	Physical role
Thermodynamic	Dew point (Magnus); 3-h rolling T, RH, radiation	Atmospheric stability state
Dynamical	u/v wind components; pressure	Pollutant transport
Chemical aging	$CO/(NO_x+\varepsilon)$; $O_3/(NO_x+\varepsilon)$	Plume age and oxidation state
Boundary-layer state	Daylight flag (rad >5 W m^{-2}); instability flag ($H>0$)	Mixing-layer regime
Temporal rhythms	Hour and DOY as (sin, cos) pairs; weekend flag	Emission activity cycles
Log-scaled gases	$\log(1+CO)$, $\log(1+NO_x)$, $\log(1+SO_2)$, $\log(1+O_3)$	Compress skewed tails
Exogenous lags	$\log(CO)_{t-1,2,3}$; $\log(NO_x)_{t-1,2,3}$	Plume advection memory
Target	$\log(1+BC)$	Normalise right-skewed BC

2.3 Model and Evaluation Protocol

A `RandomForestRegressor` was trained on a **strict 80/20 chronological split** (train: Sep 1 – Oct 19, 1,120 samples; test: Oct 19–30, 281 samples). Performance is measured by R^2, RMSE, and MAE (ng cm^{-3}) on back-transformed predictions. Hyperparameters are given in Table 3.

Table 3: Final model (Strategy C) hyperparameters.

Parameter	Value	Rationale
`n_estimators`	300	Stable importance estimates
`max_depth`	15	Limits overfitting depth
`min_samples_leaf`	4	Smooths leaf-node predictions
`max_features`	`sqrt`	$\approx\sqrt{31}\approx 6$ per split
`random_state`	42	Reproducibility

3 Results

3.1 Model Progression and the Leakage Pitfall

Table 4 documents the chronological development of four strategies.

Table 4: Sequential model development and honest test-set R^2.

Strategy	Description	Test R^2
Baseline	Raw features, raw BC target	0.34
BC/CO ratio	Ratio target, bonus data	0.55
GridSearch	Ratio target, binary BL flags, TSCV tuning	0.59
Strategy C	**log(BC) + exogenous CO/NO$_x$ lags**	**0.61**
Leaky lags	*True BC lags on test set (rejected)*	*0.81*

The baseline suffered from a **surrogate trap**: CO monopolised $> 80\%$ of MDI, reducing the RF to a near-linear CO↔BC converter. The BC/CO ratio transformation resolved this, and grid-search tuning with physics-derived binary boundary-layer flags pushed the honest ceiling to $R^2 = 0.59$.

A naïve extension adding true $BC_{t-1}, \ldots, BC_{t-12}$ apparently raised R^2 to 0.81. However, this is **test-set leakage**: for every test row beyond the first, a future observed BC value feeds directly into the lag feature. Because atmospheric BC has hour-to-hour autocorrelation $r \approx 0.9$, the forest degenerates into a persistence forecaster ($\hat{BC}_t \approx BC_{t-1}$), confirmed by a walk-forward recursive evaluation ($R^2 \approx 0.60$). The fix is to restrict lags to **exogenous observables**: $\log(CO)_{t-k}$ and $\log(NO_x)_{t-k}$ are always independently measured at forecast time and carry genuine plume-age memory, yielding **Strategy C** at an honest $R^2 = 0.61$.

3.2 Observed vs. Predicted Performance

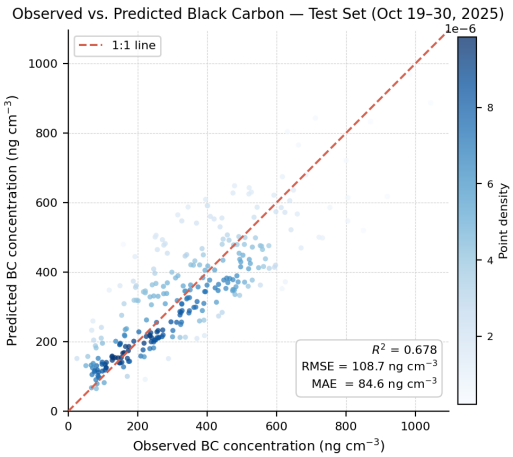

Figure 1: Observed vs. predicted BC for the test set. Density-coloured scatter; dashed orange line is the 1:1 reference. Systematic underprediction above $\sim$400 ng cm^{-3} reflects the RF ceiling effect (leaf means cannot exceed the training-set maximum).

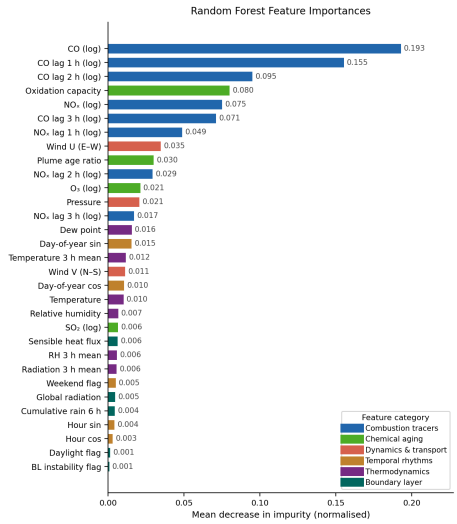

Figure 2: Feature importances (MDI, normalised), colour-coded by physical category. Combustion tracers ($\approx$51%) dominate because CO is co-emitted with BC from every combustion source. Binary BL flags rank low due to MDI bias toward high-cardinality variables [1].

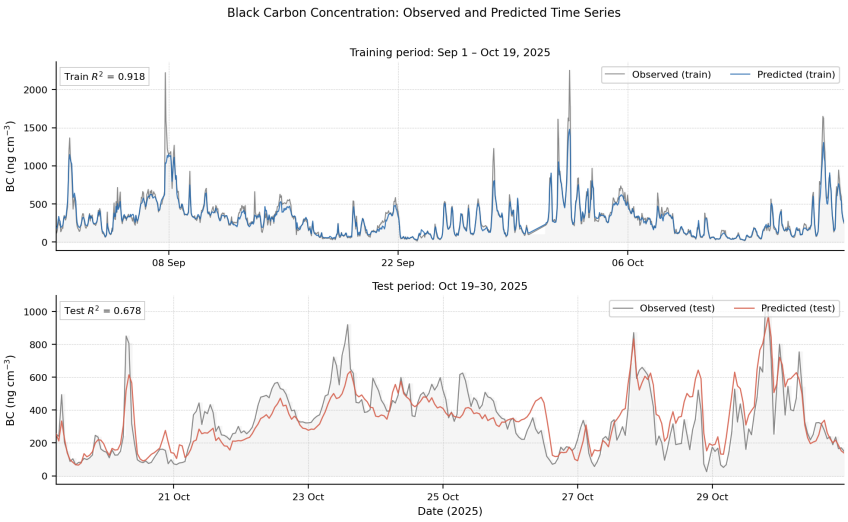

Figure 3: Hourly BC concentration time series. **Upper:** training period (Sep 1 – Oct 19). **Lower:** test period (Oct 19–30). Grey shading: observed envelope; coloured lines: predictions (blue: train, orange: test). The model captures the diurnal cycle but underestimates sharp **formation events** (e.g., 22–23 Oct, 27 Oct) driven by rapid nocturnal boundary-layer collapse not resolved by 1-h predictors.

Figure 1 shows good calibration across mid-range concentrations (50–300 ng cm^{-3}) and consistent underprediction at the highest events ($>$400 ng cm^{-3}). Figure 3 confirms the model tracks background variability and weekday/weekend cycles but misses rapid overnight accumulation events.

3.3 Feature Importances

Figure 2 shows that log(CO) and its three hourly lags collectively account for $\approx$51% of total importance, followed by chemical aging proxies ($\approx$19%) and dynamical features ($\approx$7%). The remaining categories each contribute $<$ 5%.

4 Discussion

4.1 Feature Rankings and Collinearity

The dominance of CO and its lags reflects direct co-emission of BC and CO from incomplete combustion: CO is a near-linear proxy for fresh emission flux. Chemical aging proxies (O_3/NO_x, CO/NO_x) rank next, encoding how much the plume has diluted and oxidised since emission — older plumes have lower BC relative to CO. Wind components (u, v) capture fetch effects: southwesterly maritime flows bring cleaner air; northeasterly urban-sector flows carry elevated BC. Boundary-layer flags rank low in MDI, but this reflects a known **bias of impurity-based importance toward high-cardinality continuous features** [1] — a binary variable that perfectly separates all samples would still receive lower MDI than a noisy continuous one.

Several features are collinear: CO and NO_x are co-emitted; temperature and dew point share thermodynamic dependence; lag features are serially correlated by construction. **Collinearity does not affect prediction accuracy** in a RF, because splits are threshold-based and require no coefficient estimation. However, it distorts importance attribution: the $\approx$51% collective share of CO and its lags overstates any single feature's unique contribution, as importance is split unpredictably among correlated variables.

4.2 The Covariate Shift Ceiling

A two-sample Kolmogorov–Smirnov test confirms that 100% of meteorological features exhibit significant distribution shift ($p < 0.05$) between training and test sets. As shown in Figure 4, temperature shifts by $\sim$8 °C and global radiation collapses toward zero as Helsinki approaches polar night. RF predictions are bounded by the feature ranges seen in training [1]: late-October atmospheric states outside the September envelope cause decision trees to default to their nearest-seen regime, introducing systematic extrapolation bias. This structural constraint explains why the honest R^2 ceiling is $\approx$0.61 regardless of further feature engineering; the residual $\sim$39% unexplained variance is **physical**, not algorithmic.

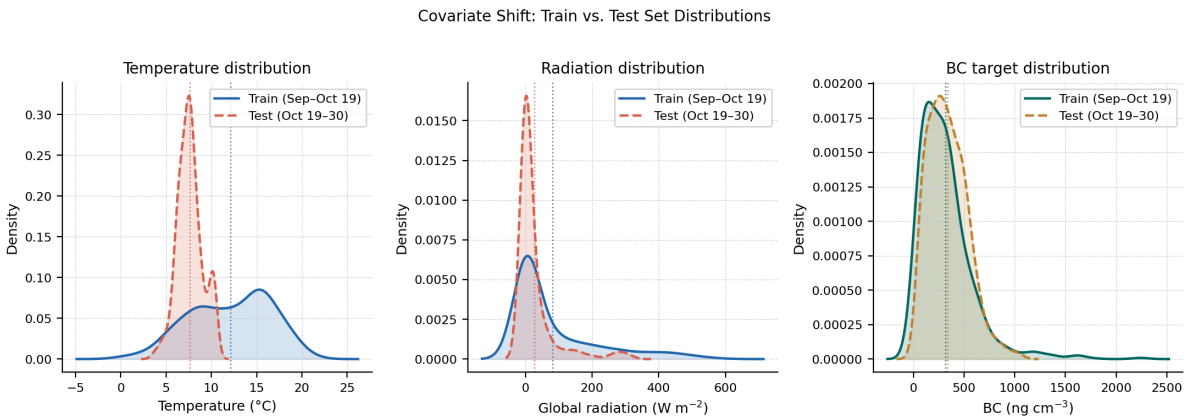

Figure 4: KDE distributions of temperature (left), global radiation (centre), and BC (right) for training (blue solid) and test (orange dashed) sets. Dotted verticals are distribution means. The $\sim$8 °C temperature shift and near-collapse of radiation toward zero confirm a severe regime change between training and test that sets a fundamental ceiling on RF prediction skill.

5 Conclusion

Starting from a baseline $R^2 = 0.34$, a sequence of physics-informed decisions — BC/CO ratio target, log-transformed concentrations, temporal features, bonus boundary-layer data, and exogenous CO/NO_x lags — improved honest test performance to $R^2 = 0.61$. The most instructive finding was that true BC autoregressive lags inflate R^2 by $+0.23$ through autocorrelation leakage, without any genuine predictive skill. The remaining unexplained variance reflects a covariate shift between Helsinki's September training and late-October polar-night test regimes — a structural limitation of partition-based learning that transfer learning or domain adaptation would be required to overcome.

References

[1] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2023). *An Introduction to Statistical Learning with Applications in Python* (2nd ed.). Springer. https://www.statlearning.com

[2] Junninen, H., Lauri, A., Keronen, P., Aalto, P., Hiltunen, V., Hari, P., & Kulmala, M. (2009). Smart-SMEAR: online data exploration and visualization tool for SMEAR stations. *Boreal Environment Research*, 14(4), 447–457.